

ASSIGNMENT – 3

Objective: Use Subqueries, CTEs, and Window Functions to analyze sales data from the Superstore dataset.

Step 1: Setup Data

- Importing data in the table “superstore_raw”.
- Create tables “Customers”, “orders”, “products” and Inserting values.

```
-- Customers table
13 CREATE TABLE customers
14 (
15     Customer_ID VARCHAR(20),
16     Customer_Name VARCHAR(100),
17     Segment VARCHAR(50)
18 );
19
20
21 INSERT INTO customers
22 SELECT DISTINCT
23     Customer_ID,
24     Customer_Name,
25     Segment
26 FROM superstore_raw;
27
28 -- Products Table
29 CREATE TABLE products
30 (
31     Product_ID VARCHAR(50),
32     Product_Name VARCHAR(255),
33     Category VARCHAR(100),
34     Sub_Category VARCHAR(100)
35 );
36
37 INSERT INTO products
38 SELECT DISTINCT
39     Product_ID,
40     Product_Name,
41     Category,
42     Sub_Category
43 FROM superstore_raw;
44
```

```
-- Orders table
45 CREATE TABLE orders
46 (
47     Order_ID VARCHAR(30),
48     Order_Date DATE,
49     Customer_ID VARCHAR(20),
50     Product_ID VARCHAR(50),
51     Sales DECIMAL(18,2),
52     Quantity INT,
53     Discount DECIMAL(5,2),
54     Profit DECIMAL(18,2)
55 );
56
57
58 INSERT INTO orders
59 SELECT DISTINCT
60     Order_ID,
61     Order_Date,
62     Customer_ID,
63     Product_ID,
64     Sales,
65     Quantity,
66     Discount,
67     Profit
68 FROM superstore_raw;
69
70
```

Step 2: Perform Subqueries and Window functions

1. Find all orders where sales are greater than the average sales.

```
79
80 -- Query 1: Find orders whose sales are greater than average sales
81 SELECT *
82 FROM orders
83 WHERE Sales >
84 (
85     -- Calculate average sales of all orders
86     SELECT AVG(Sales)
87     FROM orders
88 );
```

90 % No issues found

	Order_ID	Order_Date	Customer_ID	Product_ID	Sales	Quantity	Discount	Profit
1	CA-2015-153381	2015-09-24	DE-13255	FUR-CH-10000988	1408.10	10	0.00	394.27
2	CA-2015-153388	2015-08-01	PC-19000	FUR-TA-10003715	1004.98	6	0.20	-175.87
3	CA-2015-153416	2015-11-24	TS-21340	OFF-ST-10000060	454.86	7	0.00	54.58
4	CA-2015-153423	2015-06-22	SW-20455	OFF-BI-10000545	1217.57	2	0.20	456.59
5	CA-2015-153549	2015-03-29	SL-20155	FUR-CH-10004086	1166.92	5	0.20	131.28
6	CA-2015-153794	2015-09-11	SB-20290	OFF-AR-10000380	265.86	7	0.00	79.76
7	CA-2015-153878	2015-04-25	TS-21655	OFF-AP-10001205	272.40	5	0.00	76.27
8	CA-2015-154284	2015-12-21	SZ-20035	TEC-AC-10003198	637.44	8	0.20	135.46
9	CA-2015-154284	2015-12-21	SZ-20035	TEC-MA-10004241	600.53	2	0.30	137.26
10	CA-2015-154291	2015-07-25	YS-21880	FUR-TA-10002041	358.58	2	0.00	39.44
11	CA-2015-154326	2015-02-15	RP-19855	TEC-PH-10000560	699.98	2	0.00	195.99
12	CA-2015-154620	2015-12-12	LT-17110	FUR-CH-10004675	348.93	2	0.20	34.89
13	CA-2015-154746	2015-11-14	PJ-18835	OFF-AP-10003057	826.11	3	0.00	322.18
14	CA-2015-154823	2015-08-21	KN-16390	FUR-CH-10003379	598.46	3	0.30	-42.75
15	CA-2015-154956	2015-07-04	IM-15070	TEC-PH-10004165	1099.96	4	0.00	285.99
16	CA-2015-155306	2015-04-17	GA-14515	OFF-AP-10002651	1447.65	5	0.00	419.82
17	CA-2015-155306	2015-04-17	GA-14515	TEC-AC-10003174	222.27	2	0.00	120.25

2. Find the highest sales order for each customer.

```
90
91 -- Query 2: Find the highest sales order for every customer
92 SELECT *
93 FROM orders o
94 WHERE Sales =
95 (
96     -- Find maximum sales for the current customer
97     SELECT MAX(Sales)
98     FROM orders
99     WHERE Customer_ID = o.Customer_ID
100 );
```

90 % No issues found

	Order_ID	Order_Date	Customer_ID	Product_ID	Sales	Quantity	Discount	Profit
1	CA-2015-153381	2015-09-24	DE-13255	FUR-CH-10000988	1408.10	10	0.00	394.27
2	CA-2015-153388	2015-08-01	PC-19000	FUR-TA-10003715	1004.98	6	0.20	-175.87
3	CA-2015-153416	2015-11-24	TS-21340	OFF-ST-10000060	454.86	7	0.00	54.58
4	CA-2015-153423	2015-06-22	SW-20455	OFF-BI-10000545	1217.57	2	0.20	456.59
5	CA-2015-154284	2015-12-21	SZ-20035	TEC-AC-10003198	637.44	8	0.20	135.46
6	CA-2015-154326	2015-02-15	RP-19855	TEC-PH-10000560	699.98	2	0.00	195.99
7	CA-2015-154746	2015-11-14	PJ-18835	OFF-AP-10003057	826.11	3	0.00	322.18
8	CA-2015-154823	2015-08-21	KN-16390	FUR-CH-10003379	598.46	3	0.30	-42.75
9	CA-2015-155306	2015-04-17	GA-14515	OFF-AP-10002651	1447.65	5	0.00	419.82
10	CA-2015-155600	2015-12-04	RO-19780	OFF-BI-10000545	1598.06	7	0.70	-1065.37
11	CA-2015-155761	2015-12-11	SC-20800	TEC-AC-10001606	159.98	2	0.20	36.00
12	CA-2015-156104	2015-12-06	NP-18685	TEC-CO-10002095	999.98	2	0.00	449.99
13	CA-2015-156566	2015-10-01	EM-13960	TEC-PH-10001817	572.80	2	0.20	50.12
14	CA-2015-156734	2015-06-18	BV-11245	FUR-CH-10002965	643.14	4	0.20	56.27
15	CA-2015-157084	2015-12-19	JG-15160	TEC-AC-10002049	1265.85	3	0.00	556.97
16	CA-2015-157322	2015-07-02	RH-19600	OFF-ST-10003208	435.50	3	0.20	48.99
17	CA-2015-158421	2015-09-21	GR-14575	FUR-CH-10000200	1600.04	4	0.00	422.51

3. Calculate total sales for each customer.

```
102 -- Query 3: Calculate total sales for each customer using CTE
103
104 WITH CustomerSales AS
105 (
106     -- Summing sales customer-wise
107     SELECT Customer_ID,
108            SUM(Sales) AS TotalSales
109     FROM orders
110     GROUP BY Customer_ID
111 )
112
113 -- Display total sales of each customer
114 SELECT *
115 FROM CustomerSales;
```

90 % No issues found

	Customer_ID	TotalSales
1	LF-17185	3930.52
2	KH-16330	3312.87
3	BF-11020	2888.50
4	AH-10585	738.50
5	DW-13195	1080.76
6	SV-20785	4105.32
7	SC-20305	3979.06
8	CD-11980	639.76
9	CJ-11875	16.52
10	DK-12985	1687.03
11	HR-14830	5248.80
12	AG-10525	1455.04
13	RB-19360	15117.35
14	FP-14320	4046.75
15	JM-15655	864.95
16	JL-15505	3173.87
17	UA-14005	2162.22

4. Find customers whose total sales are above average.

```
118 -- Query 4: Find customers whose total sales are above average
119
120 WITH CustomerSales AS
121 (
122     -- Calculate total sales for each customer
123     SELECT Customer_ID,
124            SUM(Sales) AS TotalSales
125     FROM orders
126     GROUP BY Customer_ID
127 )
128
129 SELECT *
130 FROM CustomerSales
131 WHERE TotalSales >
132 (
133     -- Calculate average of total customer sales
134     SELECT AVG(TotalSales)
135     FROM CustomerSales
136 );
```

90 % No issues found

	Customer_ID	TotalSales
1	LF-17185	3930.52
2	KH-16330	3312.87
3	SV-20785	4105.32
4	SC-20305	3979.06
5	HR-14830	5248.80
6	RB-19360	15117.35
7	FP-14320	4046.75
8	JL-15505	3173.87
9	SR-20740	4345.88
10	JM-15250	3159.11
11	DG-13300	3195.82
12	MC-17845	3805.71
13	SC-20260	3729.80
14	KW-16435	4416.52
15	PM-19135	4115.66
16	SN-20710	3127.97

5. Rank all customers based on total sales.

```

139  -- Query 5: Rank customers based on their total sales
140
141  WITH CustomerSales AS
142  (
143      SELECT Customer_ID,
144             SUM(Sales) AS TotalSales
145      FROM orders
146      GROUP BY Customer_ID
147  )
148
149  SELECT Customer_ID,
150         TotalSales,
151
152         -- Assign rank based on highest sales
153         RANK() OVER(ORDER BY TotalSales DESC) AS SalesRank
154  FROM CustomerSales;

```

90 % No issues found

	Customer_ID	TotalSales	SalesRank
1	SM-20320	25043.07	1
2	TC-20980	19052.22	2
3	RB-19360	15117.35	3
4	TA-21385	14595.62	4
5	AB-10105	14473.57	5
6	KL-16645	14175.23	6
7	SC-20095	14142.34	7
8	HL-15040	12873.30	8
9	SE-20110	12209.44	9
10	CC-12370	12129.08	10
11	TS-21370	11891.75	11
12	GT-14710	11820.12	12
13	BM-11140	11789.64	13
14	SV-20365	11470.94	14
15	CJ-12010	11164.97	15
16	CL-12565	10880.56	16
17	ME-17220	10662.72	17

6. Assign row numbers to each order within a customer.

```

156
157  -- Query 6: Assign row numbers to orders for each customer
158
159  SELECT Order_ID,
160         Customer_ID,
161         Order_Date,
162
163         -- Number orders separately for each customer
164         ROW_NUMBER() OVER
165         (
166             PARTITION BY Customer_ID
167             ORDER BY Order_Date
168         ) AS RowNum
169
170  FROM orders;

```

90 % No issues found

	Order_ID	Customer_ID	Order_Date	RowNum
1	CA-2014-128055	AA-10315	2014-03-31	1
2	CA-2014-128055	AA-10315	2014-03-31	2
3	CA-2014-138100	AA-10315	2014-09-15	3
4	CA-2014-138100	AA-10315	2014-09-15	4
5	CA-2015-121391	AA-10315	2015-10-04	5
6	CA-2016-103982	AA-10315	2016-03-03	6
7	CA-2016-103982	AA-10315	2016-03-03	7
8	CA-2016-103982	AA-10315	2016-03-03	8
9	CA-2016-103982	AA-10315	2016-03-03	9
10	CA-2017-147039	AA-10315	2017-06-29	10
11	CA-2017-147039	AA-10315	2017-06-29	11
12	CA-2014-158064	AA-10375	2014-04-21	1
13	CA-2014-130729	AA-10375	2014-10-24	2
14	CA-2015-140921	AA-10375	2015-02-03	3
15	CA-2015-140921	AA-10375	2015-02-03	4
16	CA-2015-109939	AA-10375	2015-05-08	5
17	CA-2015-114502	AA-10375	2015-11-12	6

7. Display top 3 customers based on total sales.

```
174
175 -- Query 7: Display top 3 customers based on total sales
176
177 WITH CustomerSales AS
178 (
179     SELECT Customer_ID,
180            SUM(Sales) AS TotalSales
181     FROM orders
182     GROUP BY Customer_ID
183 )
184
185 SELECT *
186 FROM
187 (
188     SELECT Customer_ID,
189            TotalSales,
190
191            -- Rank customers by sales
192            RANK() OVER(ORDER BY TotalSales DESC) AS SalesRank
193
194     FROM CustomerSales
195 ) RankedCustomers
196
197 -- Show only top 3 ranked customers
198 WHERE SalesRank <= 3;
```

70 % No issues found

Results Messages

	Order_ID	Customer_ID	Order_Date	RowNum
1	CA-2014-128055	AA-10315	2014-03-31	1
2	CA-2014-128055	AA-10315	2014-03-31	2
3	CA-2014-138100	AA-10315	2014-09-15	3
4	CA-2014-138100	AA-10315	2014-09-15	4
5	CA-2015-121391	AA-10315	2015-10-04	5
6	CA-2016-103982	AA-10315	2016-03-03	6
7	CA-2016-103982	AA-10315	2016-03-03	7
8	CA-2016-103982	AA-10315	2016-03-03	8
9	CA-2016-103982	AA-10315	2016-03-03	9
10	CA-2017-147039	AA-10315	2017-06-29	10
11	CA-2017-147039	AA-10315	2017-06-29	11
12	CA-2014-158064	AA-10375	2014-04-21	1
13	CA-2014-130729	AA-10375	2014-10-24	2
14	CA-2015-140921	AA-10375	2015-02-03	3
15	CA-2015-140921	AA-10375	2015-02-03	4
16	CA-2015-109939	AA-10375	2015-05-08	5

Step 3: Final Combined Query Write one final query that shows:

- Customer Name
- Total Sales
- Rank

```
201
202 -- STEP 3. Final Query: Display customer name, total sales and rank
203
204 WITH CustomerSales AS
205 (
206     -- Calculate total sales for each customer
207     SELECT Customer_ID,
208            SUM(Sales) AS TotalSales
209     FROM orders
210     GROUP BY Customer_ID
211 )
212
213 SELECT
214     c.Customer_Name,
215     cs.TotalSales,
216
217     -- Rank customers based on total sales
218     RANK() OVER(ORDER BY cs.TotalSales DESC) AS SalesRank
219
220 FROM CustomerSales cs
221 JOIN customers c
222     ON cs.Customer_ID = c.Customer_ID;
```

70 %

✓ No issues found

Results Messages

	Customer_Name	TotalSales	SalesRank
1	Sean Miller	25043.07	1
2	Tamara Chand	19052.22	2
3	Raymond Buch	15117.35	3
4	Tom Ashbrook	14595.62	4
5	Adrian Barton	14473.57	5
6	Ken Lonsdale	14175.23	6
7	Sanjit Chand	14142.34	7
8	Hunter Lopez	12873.30	8
9	Sanjit Engle	12209.44	9
10	Christopher Conant	12129.08	10
11	Todd Sumrall	11891.75	11
12	Greg Tran	11820.12	12
13	Becky Martin	11789.64	13
14	Seth Vernon	11470.94	14
15	Caroline Jumper	11164.97	15
16	Clay Ludtke	10880.56	16

Mini Project: Customer Sales Insights

1. Who are the top 5 customers?

```
227 -- MINI PROJECT QUESTIONS
228 -- QUES 1. TOP 5 CUSTOMERS
229
230 WITH customer_sales AS
231 (
232     SELECT customer_id,
233            SUM(sales) AS total_sales
234     FROM orders
235     GROUP BY customer_id
236 )
237 SELECT TOP 5
238     c.customer_name,
239     cs.total_sales
240 FROM customer_sales cs
241 JOIN customers c
242 ON cs.customer_id = c.customer_id
243 ORDER BY cs.total_sales DESC;
```

70 % No issues found

	customer_name	total_sales
1	Sean Miller	25043.07
2	Tamara Chand	19052.22
3	Raymond Buch	15117.35
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.57

2. Who are the bottom 5 customers?

```
246 -- QUES 2. BOTTOM 5 CUSTOMERS
247 WITH customer_sales AS
248 (
249     SELECT customer_id,
250            SUM(sales) AS total_sales
251     FROM orders
252     GROUP BY customer_id
253 )
254 SELECT TOP 5
255     c.customer_name,
256     cs.total_sales
257 FROM customer_sales cs
258 JOIN customers c
259 ON cs.customer_id = c.customer_id
260 ORDER BY cs.total_sales ASC;
```

100 % No issues found

	customer_name	total_sales
1	Thais Sissman	4.84
2	Lela Donovan	5.30
3	Carl Jackson	16.52
4	Mitch Gastineau	16.74
5	Roy Skaria	22.33

3. Which customers made only one order?

```
262
263 -- QUES 3. CUSTOMERS WHO MADE ONLY ONE ORDER
264
265 SELECT
266     c.customer_name,
267     COUNT(o.order_id) AS order_count
268 FROM customers c
269 JOIN orders o
270 ON c.customer_id = o.customer_id
271 GROUP BY c.customer_name
272 HAVING COUNT(o.order_id) = 1;
```

100 % No issues found

	customer_name	order_count
1	Anthony O'Donnell	1
2	Carl Jackson	1
3	Jocasta Rupert	1
4	Lela Donovan	1
5	Ricardo Emerson	1

4. Which customers have above-average sales?

```
274 */
275 -- QUES 4. Customers With Above-Average Sales
276
277 WITH customer_sales AS
278 (
279     SELECT customer_id,
280            SUM(sales) AS total_sales
281     FROM orders
282     GROUP BY customer_id
283 )
284 SELECT *
285 FROM customer_sales
286 WHERE total_sales >
287 (
288     SELECT AVG(total_sales)
289     FROM customer_sales
290 );
```

100 % No issues found

	customer_id	total_sales
1	LF-17185	3930.52
2	KH-16330	3312.87
3	SV-20785	4105.32
4	SC-20305	3979.06
5	HR-14830	5248.80
6	RB-19360	15117.35
7	FP-14320	4046.75
8	JL-15505	3173.87
9	SR-20740	4345.88
10	JM-15250	3159.11
11	DG-13300	3195.82
12	MC-17845	3805.71
13	SC-20260	3729.80
14	KW-16435	4416.52
15	PM-19135	4115.66
16	SN-20710	3127.97
17	XS-21880	8720.44

5. What is the highest order value per customer?

293	-- QUES 5. Highest Order Value Per Customer
294	
295	SELECT
296	customer_id,
297	MAX(sales) AS highest_order_value
298	FROM orders
299	GROUP BY customer_id;
100 % No issues found	
Results Messages	
	customer_id highest_order_value
1	LF-17185 2177.58
2	KH-16330 833.94
3	BF-11020 1035.80
4	AH-10585 383.96
5	DW-13195 318.43
6	SV-20785 844.12
7	SC-20305 2803.92
8	CD-11980 318.96
9	CJ-11875 16.52
10	DK-12985 646.20
11	HR-14830 2399.96
12	AG-10525 559.71
13	RB-19360 13999.96
14	FP-14320 1317.49
15	JM-15655 383.95
16	JL-15505 1473.10
17	HA-14905 579.14
18	JD-16060 221.16
19	EA-14035 599.99
20	SR-20740 3404.50
21	JM-15580 105.55
22	JM-15250 908.82
23	DG-13300 1979.89
24	MC-17845 1669.60
25	EB-14170 807.75
26	ER-13855 876.30
27	SC-20260 1552.83